

Policy Recommendation: Structuring Insurance Coverage for High-Value Cancer Treatment

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Summary

This brief evaluates whether an insurance product should cover a cancer treatment based on patient preferences estimated from a discrete choice experiment. Using an individual fixed-effects regression model, the analysis estimates how treatment characteristics such as regimen type, visit frequency, overall survival, and out-of-pocket cost influence patient choice.

The results suggest that patients place meaningful value on treatment regimen and overall survival. Regimen 2 and regimen 4 were preferred relative to the baseline regimen, while regimen 3 was less preferred. Overall survival was also positively associated with treatment choice. Visit frequency and out-of-pocket cost were not statistically significant in the model.

Based on the willingness-to-pay framework, patients appear to value certain treatment improvements above an actuarially fair insurance premium. However, because the out-of-pocket cost coefficient was small, positive, and statistically insignificant, the exact willingness-to-pay estimates should be interpreted cautiously. The recommended policy is to offer coverage for the treatment, but not through a fully actuarially fair premium alone. Instead, the insurer should use a structured benefit design that includes premiums, cost-sharing, and risk-pooling protections to reduce moral hazard and adverse selection.

Background

Cancer treatment decisions often involve trade-offs between clinical benefits, treatment burden, and financial cost. Patients may prefer treatments that offer longer survival or more favorable regimens, but insurers must decide whether coverage is financially sustainable. Health economics provides tools for evaluating these trade-offs by estimating how much patients value specific treatment attributes.

This analysis uses a discrete choice experiment, where individuals are assumed to choose among treatment options based on the utility they receive from each option's attributes. The regression model estimates the relationship between treatment attributes and the likelihood that a treatment is chosen.

The model used was:

choice = regimen indicators + visit frequency + overall survival + out-of-pocket cost + individual fixed effects

The individual fixed-effects approach controls for stable differences across respondents, such as general risk tolerance, health preferences, or baseline attitudes toward medical treatment. This

allows the analysis to focus more directly on how changes in treatment attributes affect choices within individuals.

Regression Results and Interpretation

The regression estimated the following coefficients:

Variable	Coefficient	Standard Error	Statistical Significance	Interpretation
Regimen 2	0.313	0.024	***	Regimen 2 was preferred relative to Regimen 1, the baseline regimen.
Regimen 3	-0.062	0.024	***	Regimen 3 was less preferred relative to Regimen 1.
Regimen 4	0.181	0.024	***	Regimen 4 was preferred relative to Regimen 1.
Visit Frequency	-0.005	0.005	Not significant	More frequent visits may reduce preference, but the effect was not statistically significant.
Overall Survival	0.007	0.002	***	Longer overall survival increased the likelihood that a treatment was chosen.
Out-of-Pocket Cost	0.00005	0.0001	Not significant	Out-of-pocket cost was not statistically significant in this model.

Model notes: Individual fixed-effects regression; dependent variable = treatment choice. Regimen 1 is the omitted baseline category. Significance levels: *** $p < 0.01$.

Regimen 1 served as the baseline category. The positive coefficient on regimen 2 suggests that patients preferred regimen 2 relative to regimen 1. Regimen 4 was also preferred relative to the baseline, although less strongly than regimen 2. Regimen 3 had a negative coefficient, suggesting it was less preferred than regimen 1.

Overall survival had a positive and statistically significant coefficient, indicating that patients were more likely to choose treatments that offered additional survival benefit. This is consistent with expectations in a cancer treatment context, where longer survival is a central measure of treatment value.

Visit frequency was negative but not statistically significant, suggesting that more frequent visits may reduce treatment attractiveness, but the model does not provide strong evidence of that

relationship. Out-of-pocket cost was also not statistically significant. This is important because willingness-to-pay calculations rely heavily on the cost coefficient.

Willingness-to-Pay Framework

In a discrete choice experiment, willingness to pay is calculated by comparing the value of a treatment attribute to the cost coefficient. The general formula is:

$$\text{WTP} = - \text{attribute coefficient} / \text{cost coefficient}$$

The cost coefficient acts like an exchange rate between utility and dollars. If a treatment attribute increases utility, and higher cost lowers utility, then dividing the attribute coefficient by the cost coefficient gives the implied dollar amount patients would be willing to pay for that attribute.

In this case, the original calculation added multiple regimen coefficients together. A more appropriate approach is to calculate willingness to pay for each mutually exclusive regimen separately or to calculate willingness to pay for one clearly defined treatment profile. Because a treatment cannot simultaneously be regimen 2, regimen 3, and regimen 4, those coefficients should not all be added together unless the assignment explicitly defines them as cumulative attributes rather than mutually exclusive categories.

Using the model estimates, the implied willingness to pay for individual attributes would be:

- Regimen 2 relative to regimen 1: approximately \$6,260
- Regimen 3 relative to regimen 1: approximately -\$1,240
- Regimen 4 relative to regimen 1: approximately \$3,620
- 24 months of additional overall survival: approximately \$3,360

These estimates suggest that patients place the highest value on regimen 2 among the regimen options and also value additional survival. A combined treatment profile could be evaluated by adding the relevant regimen coefficient and the survival benefit, then dividing by the cost coefficient. For example, a treatment with regimen 2 and 24 additional months of survival would have a higher implied value than regimen 2 alone.

However, these willingness-to-pay estimates should be interpreted with caution because the out-of-pocket cost coefficient was small, positive, and statistically insignificant. In theory, higher out-of-pocket costs should reduce the probability that a treatment is chosen. The unexpected sign and lack of statistical significance mean that the model does not provide a strong or stable estimate of price sensitivity.

Insurance Premium Analysis

An actuarially fair insurance premium is a premium equal to the insurer's expected payout. Under the assumptions used in the original analysis, the lifetime probability of cancer was

estimated at 40.5%, average life expectancy was assumed to be 80 years, and one year of life was valued at \$100,000. Dividing the lifetime cancer probability across 80 years gives an approximate annual cancer risk of 0.5%.

The expected annual cost is therefore:

$$0.005 \times \$100,000 = \$500$$

This implies an actuarially fair premium of approximately \$500 per year, or \$1,000 over 24 months. Compared with the willingness-to-pay estimates, this premium appears affordable relative to the value patients place on certain treatment benefits, especially treatments involving preferred regimens or additional survival.

From a policy perspective, this suggests that there may be economic justification for covering the treatment. Patients appear to value the treatment benefits more than the expected actuarially fair cost. However, actuarial fairness alone is not enough to determine whether the insurance product is sustainable.

Policy Concerns

Two major insurance market problems should be considered: moral hazard and adverse selection.

Moral hazard occurs when insurance changes behavior because individuals no longer bear the full cost of their choices. In this context, full insurance coverage could increase utilization or reduce incentives to avoid health risks. While the extent of moral hazard in cancer treatment may be limited compared with other forms of care, it remains relevant for benefit design.

Adverse selection is a more serious concern. Individuals with a family history of cancer or private information about their health risks may be more likely to purchase the insurance. If higher-risk individuals disproportionately enroll, the insurer's average cost will rise. Premiums may then increase, causing healthier individuals to leave the insurance pool. This can lead to a cycle of rising premiums and worsening risk composition.

Risk aversion supports the existence of insurance. People are generally willing to pay a certain premium to avoid the possibility of a large financial loss. This makes cancer treatment coverage valuable, especially when the potential cost of treatment is high and uncertain. Still, insurance must be designed carefully to balance financial protection with long-term sustainability.

Recommendation

The insurer should offer coverage for the cancer treatment, but should not rely on a simple actuarially fair premium with full coverage and no cost-sharing. A more sustainable policy would include:

1. A base premium that reflects expected treatment costs.

2. Moderate cost-sharing, such as copayments or coinsurance, to reduce excessive utilization.
3. Deductibles or benefit limits designed carefully so they do not prevent access to high-value care.
4. Broad enrollment incentives to reduce adverse selection.
5. Periodic premium review based on actual claims experience.
6. Clear coverage rules tied to clinical eligibility and evidence-based treatment guidelines.

This structure would preserve the main value of insurance: protecting patients from large financial losses. At the same time, it would reduce the risk that the insurer underprices the product or attracts only the highest-risk patients.

Conclusion

The discrete choice experiment suggests that patients value cancer treatment attributes such as regimen type and overall survival. The willingness-to-pay estimates indicate that certain treatment improvements may be valued above the actuarially fair premium calculated in the assignment. However, the cost coefficient in the regression was not statistically significant, so the exact dollar values should be treated as suggestive rather than definitive.

The best policy recommendation is to cover the treatment through a structured insurance product rather than a simple fully insured actuarially fair plan. A design that combines premiums, cost-sharing, and safeguards against adverse selection would better balance patient access, financial protection, and insurer sustainability.